

CITY GENERAL HOSPITAL — DEPARTMENT OF GENERAL MEDICINE
Clinical Case Report & Patient Discharge Summary

Dengue Fever with Thrombocytopenia (DENV-2 Serotype): Clinical Course, Laboratory Profile, Toxicological Screening, and Multidisciplinary Management — A Single-Patient Clinical Record

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Report ID: CGH-DR-2026-04891-0430 · Patient ID: CGH-2024-04891 · UHID: UHID-MH-2024-948271

This document constitutes the official clinical discharge record of the above-named patient and has been prepared in accordance with the National Medical Commission (NMC) guidelines, the Digital Personal Data Protection Act 2023, and the clinical documentation standards of City General Hospital.

ABSTRACT

Background: Dengue fever remains a significant public health burden in tropical regions, with Dengue Haemorrhagic Fever presenting particular challenges due to thrombocytopenia and capillary leak syndrome. This report documents the full clinical course of a 46-year-old male with secondary DENV-2 dengue infection complicated by severe thrombocytopenia (platelet nadir 38,000/ μ L) and associated transaminitis.

Case Presentation: The patient presented with a 5-day history of high-grade fever (104.1°F), severe myalgia, retro-orbital headache, and maculopapular rash. NS1 Antigen was positive on admission; secondary dengue infection was confirmed by positive IgG/IgM serology and RT-PCR identifying DENV-2 serotype. Thrombocytopenia reached a nadir of 38,000/ μ L on Day 4, necessitating prophylactic single-donor platelet transfusion. Significant transaminitis (SGPT 88 U/L, SGOT 102 U/L) and mild pleural effusion with ascites were documented. Pre-existing Type 2 Diabetes Mellitus and Hypertension were managed concurrently.

Investigations: Comprehensive laboratory evaluation including haematology, biochemistry, serology, microbiology, blood alcohol content, 10-panel drug of abuse screen, poison and toxin screen, and 12-element heavy metals panel (ICP-MS) was performed. Chest X-Ray, abdominal ultrasonography, and ECG were the primary imaging modalities utilised.

Outcome: Patient responded favourably to conservative management. Fever resolved by Day 7; platelet count recovered to 92,000/ μ L by Day 8. NEWS2 score at discharge: 0. Patient discharged in stable condition on Day 9 with structured follow-up plan.

Keywords: Dengue fever; DENV-2; thrombocytopenia; capillary leak syndrome; transaminitis; platelet transfusion; NEWS2 early warning score; ICP-MS heavy metals; clinical discharge record

1. PATIENT IDENTIFICATION & DEMOGRAPHICS

Table 1.

Patient demographics, admission details, and administrative identifiers.

Parameter	Value	Parameter	Value
Full Name	Rajesh Kumar Sharma	Patient ID	CGH-2024-04891
Date of Birth	12 March 1978 (Age: 46 years)	UHID	UHID-MH-2024-948271
Sex	Male	Blood Group	B Positive (B+), Rh Positive
Nationality	Indian	Occupation	Accountant
Marital Status	Married	Ward / Bed	General Ward — Block B2, Bed 14
Admission	22 April 2026, 09:14 hrs	Discharge	30 April 2026, 11:30 hrs
Length of Stay	8 Days	Code Status	Full Code — Resuscitate
Emergency Contact	Priya Sharma (Spouse) — +91-98765-43210	Insurance	Star Health Insurance — Policy No. SH-MH-2024-112233
TPA / Claim	Medi Assist TPA Pvt. Ltd. Claim No. MA-2026-48912	Discharge Mode	Discharged to Home (Self-Care)

1.1 Diagnosis

Admission Diagnosis: Dengue Fever with Thrombocytopenia (NS1 Antigen positive; DENV-2 serotype)

Discharge Diagnosis: Dengue Fever — Recovered. Thrombocytopenia — Resolving.

Condition at Discharge: Stable. Afebrile for 48 hours. Platelet count recovering satisfactorily.

1.2 Medical History

Category	Details
Comorbidities	Type 2 Diabetes Mellitus (on oral hypoglycaemic agents); Hypertension (on Amlodipine 5 mg)
Surgical History	Nil significant
Family History	Father: Type 2 DM; Mother: Hypertension
Social History	Non-smoker. Occasional social alcohol (last consumption 14 Apr 2026 — approximately 2 units). No recreational drug use reported.

1.3 Allergy Profile

Allergen	Category	Reaction	Severity	Verification
Penicillin / Amoxicillin	Drug	Urticaria, rash	Moderate	RAST confirmed 2019
Aspirin / NSAIDs (Group)	Drug	GI upset — CONTRAINDICATED in dengue	HIGH — Avoid	Clinical protocol
Shellfish (Prawns, Crab)	Food	Urticaria, lip angioedema	Moderate	Self-reported
Groundnuts / Peanuts	Food	Mild oral tingling	Mild	Self-reported
Dust mites	Environmental	Rhinitis, sneezing	Mild	Self-reported

Note: All documented allergies were flagged in the medication administration record (MAR) and reviewed by the treating physician and pharmacist prior to each prescription and dispensing event.

1.4 Dietary Restrictions

- Low glycaemic index diet — Diabetic diet protocol (Type 2 DM)
- Sodium-restricted diet < 2 g/day (Hypertension management)
- Minimum 3 litres fluid intake daily (Dengue fluid resuscitation protocol)
- Abstain from alcohol completely — minimum 4 weeks post-discharge (hepatotoxicity risk — elevated LFTs)

5. Avoid Aspirin, NSAIDs, Diclofenac — increased bleeding risk in dengue convalescence

2. CLINICAL PRESENTATION

2.1 History of Presenting Illness

The patient, a 46-year-old male with a background of Type 2 Diabetes Mellitus and Hypertension, presented to the outpatient department of City General Hospital on 22 April 2026 with a 5-day history of high-grade, continuous fever (103–104°F) that had responded only partially to self-administered over-the-counter Paracetamol. Accompanying symptoms included severe retro-orbital headache (rated 8/10 on the Visual Analogue Scale), generalised myalgia and arthralgia consistent with the classic 'breakbone fever' presentation, nausea with 3–4 episodes of vomiting daily for the first three days, and a maculopapular rash that developed on the trunk and bilateral upper limbs from Day 3 of illness. No haemorrhagic manifestations (gum bleeding, haemoptysis, melaena, haematuria) were reported at the time of admission. The patient reported no recent travel outside the Pune district; however, his residential locality had documented active dengue transmission activity, and a neighbour had been hospitalised with confirmed dengue 10 days prior to the index presentation.

2.2 Physical Examination at Admission — 22 April 2026, 09:30 hrs

Table 2.

Systematic physical examination findings at the time of admission, 22 April 2026.

System / Parameter	Clinical Finding
General Condition	Alert, oriented to time/place/person. Moderately ill-looking. Clinically dehydrated.
Temperature	103.8°F (39.9°C) — Axillary
Pulse	98 beats/min, regular rhythm, low volume
Blood Pressure	118/76 mmHg — Right arm, sitting position
Respiratory Rate	18 breaths/min
SpO₂	97% on room air
Icterus	Mild scleral icterus noted bilaterally
Lymphadenopathy	Mild bilateral cervical lymphadenopathy — non-tender, mobile
Skin / Rash	Maculopapular rash over trunk and bilateral upper limbs. Tourniquet test (Rumpel-Leede): Positive — 22 petechiae in 1 inch ² area (threshold > 20).
Abdomen	Liver palpable 2 cm below right costal margin — tender. Mild splenomegaly. Mild diffuse tenderness. Shifting dullness: Positive (mild ascites).
Respiratory System	Bilateral air entry equal. Mild dullness to percussion at right lung base — consistent with early pleural effusion. No wheeze or crepitations.
Cardiovascular	S1 S2 heard normally. No murmurs. No pericardial rub.
Neurological	GCS 15/15 (E4V5M6). No focal neurological deficit. No meningismus.

3. VITAL SIGNS & CLINICAL MONITORING

Vital signs were recorded every four hours throughout the admission period. The charts below present the 09:00 hrs (morning) readings for each parameter across all nine days of admission. The shaded region represents the clinically acceptable (safe) reference range for each parameter. Figure 1 presents a multi-panel overview of six core haemodynamic parameters; Figures 2 and 3 present the platelet count trend and blood sugar trend respectively, both of which were of primary clinical significance in this case.

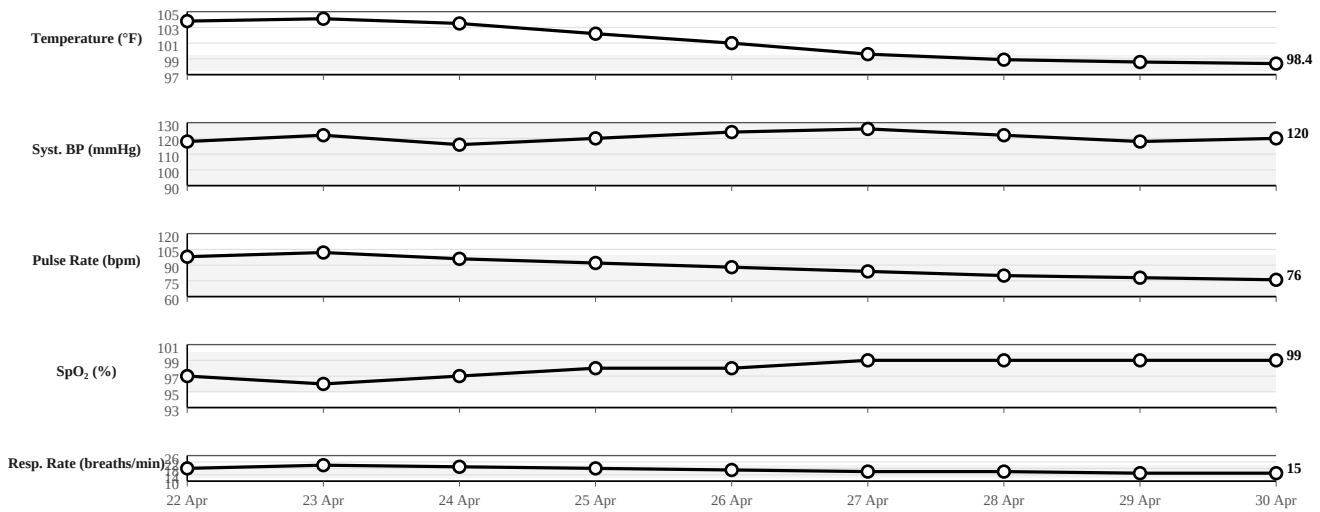


Figure 1.

Multi-panel vital signs trend across the full admission period (22–30 April 2026). Parameters shown (top to bottom): Temperature (°F), Systolic Blood Pressure (mmHg), Pulse Rate (bpm), SpO₂ (%), and Respiratory Rate (breaths/min). Shaded region = clinically safe reference range. All morning (09:00 hrs) readings shown. The progressive normalisation of all parameters from Day 5 onward is consistent with dengue fever resolution.

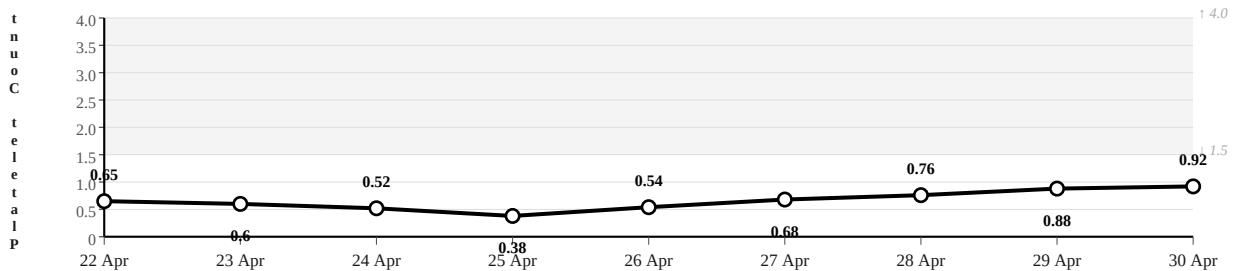


Figure 2.

Serial platelet count trend across the 9-day admission (22–30 April 2026). A progressive decline to a nadir of 0.38 lakhs/ μ L (38,000/ μ L) is observed on Day 4 (25 April 2026), consistent with dengue-associated bone marrow suppression and consumptive thrombocytopenia. A single-donor platelet transfusion was administered on Day 4. Recovery to 0.92 lakhs/ μ L (92,000/ μ L) was documented at discharge on Day 8. Shaded region = normal reference range (1.50–4.00 lakhs/ μ L).

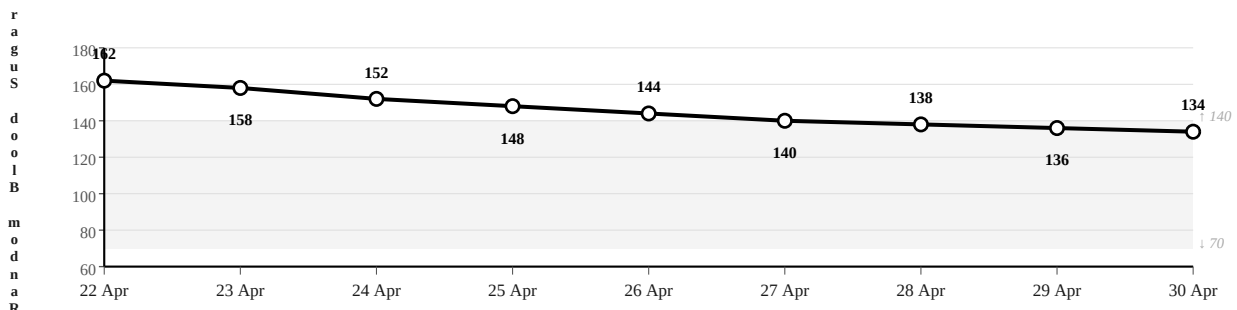
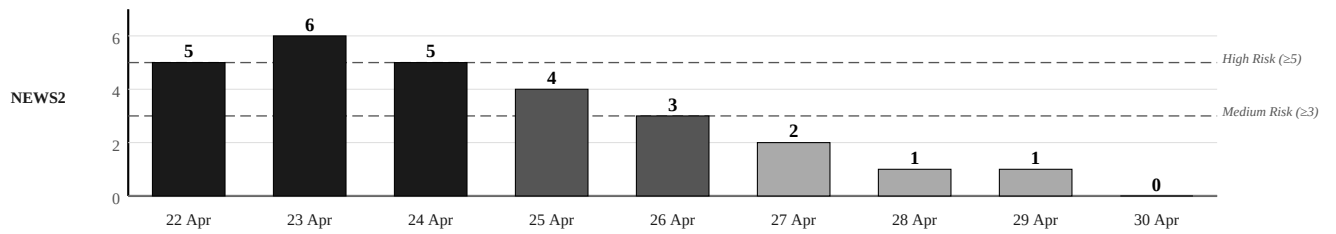


Figure 3.

Serial random blood glucose measurements across the admission period (22–30 April 2026). Consistently elevated values reflect the patient's pre-existing Type 2 Diabetes Mellitus (HbA1c 7.2%). A gradual downward trend from 162 mg/dL on Day 1 to 134 mg/dL on Day 8 is noted, attributable to improved dietary intake and ongoing Metformin therapy. Shaded region = normal fasting reference range (70–140 mg/dL).

**Figure 4.**

National Early Warning Score 2 (NEWS2) across the admission period (22–30 April 2026). The score is computed from six physiological parameters: respiratory rate, SpO_2 , systolic blood pressure, pulse rate, consciousness level, and temperature. Dashed lines indicate the medium risk threshold (score ≥ 3) and high risk threshold (score ≥ 5). Bar shading: light grey = low risk (0–2); mid grey = medium risk (3–4); dark grey = high risk (≥ 5). The patient's score declined from 6 (High Risk) on Day 2 to 0 (Low Risk) at the time of discharge, confirming clinical resolution.

4. LABORATORY INVESTIGATIONS

Comprehensive laboratory evaluation was performed at three time points: admission (Day 1, 22 April 2026), mid-stay (Day 4, 25 April 2026), and discharge (Day 8, 30 April 2026). All samples were processed by the CGH Pathology Laboratory (NABL Accreditation Certificate No. NABL-2024-00781). Reference ranges are those established for adult male subjects using the reagent kits and analysers in use at CGH at the time of analysis. The symbol '—' denotes that the test was not repeated at that time point. 'WNL' denotes Within Normal Limits.

Table 3.
Serial laboratory results at admission (Day 1, 22 Apr), mid-stay (Day 4, 25 Apr), and discharge (Day 8, 30 Apr 2026). WNL = Within Normal Limits. '—' = not repeated at that time point. Trend: ↑ = improving toward normal; ↑↑ = rapidly improving.

Investigation	Unit	Day 1	Day 4	Day 8	Reference Range	Status	Tre nd
Haemoglobin (Hb)	g/dL	13.2	12.8	13.1	13.0–17.0	WNL	—
Total WBC Count	cells/ μ L	3,200	2,800	4,100	4,000–11,000	Low → Recovering	↑
Platelet Count	lakhs/ μ L	0.65	0.38	0.92	1.50–4.00	Critically Low → Recovering	↑↑
Packed Cell Volume (PCV)	%	38.4	37.2	38.8	40–54	WNL	—
NS1 Antigen — Dengue	—	Positive	—	Negative	Negative	Positive on admission	—
Dengue IgG / IgM	—	— / —	POS / POS	POS / NEG	Negative	Secondary dengue — seroconverted	—
Dengue PCR (RT-PCR)	copies/mL	Detected (DENV-2)	—	—	Not detected	DENV-2 serotype	—
Random Blood Sugar (RBS)	mg/dL	162	148	134	70–140	Elevated → Improving	↑
Fasting Blood Sugar (FBS)	mg/dL	118	112	106	70–110	Elevated → Improving	↑
HbA1c (Glycated Hb)	%	7.2	—	—	< 5.7 Normal; > 6.4 DM	Elevated — DM diagnosis confirmed	—
Blood Urea Nitrogen (BUN)	mg/dL	22	20	18	7–25	WNL	—
Serum Creatinine	mg/dL	1.1	1.0	0.9	0.6–1.2	WNL	—
eGFR	mL/min/1.73m ²	71	74	78	> 60	WNL	—
SGPT / ALT	U/L	88	72	54	0–45	Elevated → Improving (transaminitis)	↑
SGOT / AST	U/L	102	84	61	0–40	Elevated → Improving	↑
Serum Bilirubin (Total)	mg/dL	2.1	1.8	1.3	0.2–1.2	Elevated → Improving	↑
ALP	U/L	142	128	118	44–147	Elevated → Improving	↑
Serum Albumin	g/dL	3.2	3.4	3.6	3.5–5.0	Low → Recovering	↑
Serum Sodium (Na ⁺)	mEq/L	136	138	139	136–145	WNL	—
Serum Potassium (K ⁺)	mEq/L	3.6	3.8	4.0	3.5–5.1	WNL	—
Serum Chloride (Cl ⁻)	mEq/L	102	104	103	98–107	WNL	—
Plasma Fibrinogen	mg/dL	210	195	240	200–400	Low Day 4 → Normal	—
D-Dimer	μ g/mL FEU	0.8	1.1	0.6	< 0.5	Elevated — dengue related	—
PT / INR	sec / ratio	13.2 / 1.1	13.8 / 1.15	12.9 / 1.08	11–13 / < 1.2	WNL	—
C-Reactive Protein (CRP)	mg/L	48	36	12	< 5.0	Elevated → Improving	↑
ESR	mm/hr	42	38	22	0–15	Elevated → Improving	↑

Investigation	Unit	Day 1	Day 4	Day 8	Reference Range	Status	Trend
Serum LDH	U/L	420	380	290	120–246	Elevated — dengue marker	↑
Serum Ferritin	ng/mL	680	520	310	12–300	Elevated — dengue marker	↑
HIV I & II (Screen)	—	Non-reactive	—	—	Non-reactive	Negative	—
HBsAg (Hepatitis B)	—	Negative	—	—	Negative	Negative	—
Anti-HCV (Hepatitis C)	—	Negative	—	—	Negative	Negative	—
Urine Routine & Microscopy	—	Trace protein, RBC 2–3/hpf	—	Normal	Normal	Mild proteinuria Day 1	—
Blood Culture × 2 (aerobic)	—	No growth (5 days)	—	—	No growth	Sterile	—
Stool Occult Blood	—	Negative	—	—	Negative	Negative	—

Abbreviations: Hb — Haemoglobin; WBC — White Blood Cell; PCV — Packed Cell Volume; NS1 — Non-Structural Protein 1; RT-PCR — Reverse Transcriptase Polymerase Chain Reaction; RBS — Random Blood Sugar; FBS — Fasting Blood Sugar; HbA1c — Glycated Haemoglobin; BUN — Blood Urea Nitrogen; eGFR — Estimated Glomerular Filtration Rate; SGPT/ALT — Serum Glutamate-Pyruvate Transaminase / Alanine Transaminase; SGOT/AST — Serum Glutamate-Oxaloacetate Transaminase / Aspartate Transaminase; ALP — Alkaline Phosphatase; CRP — C-Reactive Protein; ESR — Erythrocyte Sedimentation Rate; LDH — Lactate Dehydrogenase; PT/INR — Prothrombin Time/International Normalised Ratio; R/M — Routine and Microscopy.

5. TOXICOLOGY SCREEN, DRUG OF ABUSE PANEL & HEAVY METALS ANALYSIS

A comprehensive toxicological evaluation was performed on samples collected on 22 April 2026 at 09:45 hrs (urine — random voided sample; and venous blood — 5 mL EDTA tube) as part of the hospital's standard admission protocol. Analysis was conducted by the CGH Toxicology and Clinical Chemistry Laboratory (NABL Accredited). Results are detailed in Sections 5.1–5.4 below.

5.1 Blood Alcohol Content (BAC)

Parameter	Result	Method	Reference / Legal Limit
BAC — Venous Blood	< 10 mg/dL — Not detected (Method: GC-FID)	Gas Chromatography (GC-FID)	Not detected Legal limit (India, MV Act 1988): 30 mg/dL
BAC — Urine	Not detected	GC-FID	Not detected
Clinical Interpretation	No evidence of acute alcohol intoxication at time of admission. Patient reported last alcohol consumption 8 days prior to admission.	—	—

5.2 Drug of Abuse Screen — 10-Panel

Table 4.
10-panel drug of abuse screening — urine and blood. All positive screens confirmed by GC-MS.

Substance / Drug Class	Result	Analytical Method	Flag
Opioids — Morphine / Codeine / Heroin metabolites	Not Detected	GC-MS / ELISA	NEG
Cannabis / THC (Tetrahydrocannabinol)	Not Detected	GC-MS (confirmed)	NEG
Cocaine / Benzoylcegonine	Not Detected	GC-MS	NEG
Amphetamines / Methamphetamine	Not Detected	GC-MS	NEG
Benzodiazepines (group screen)	Not Detected	ELISA	NEG
Barbiturates (group screen)	Not Detected	ELISA	NEG
Phencyclidine (PCP)	Not Detected	ELISA	NEG
Tricyclic Antidepressants (group)	Not Detected	ELISA	NEG
Methadone	Not Detected	ELISA	NEG
MDMA / Ecstasy	Not Detected	GC-MS	NEG

Note: All negatives on the 10-panel screen were confirmed by Gas Chromatography — Mass Spectrometry (GC-MS) where indicated. ELISA results without confirmatory GC-MS are noted as such. Results are disclosed only to the treating physician and the patient. Third-party disclosure requires written patient consent or a valid court order in accordance with applicable Indian law, including the Narcotic Drugs and Psychotropic Substances (NDPS) Act 1985 where relevant.

5.3 Poison, Toxin & Venom Screen

Table 5.
Poison, toxin, and venom screen results. Envenomation screen was not clinically indicated.

Substance / Test	Result	Method	Status	Reference Range
Organophosphates — Serum Cholinesterase Activity	98% activity	Colorimetric	NORMAL	70–100% activity
Carbon Monoxide (COHb%)	1.2% (Normal)	Co-oximetry	NORMAL	< 3% non-smoker
Cyanide (whole blood)	Not Detected	Microdiffusion	NEG	< 0.1 mg/L
Paracetamol (serum level)	12 mg/L — Therapeutic	Immunoassay	THERAPEUTIC	10–20 mg/L
Methanol	Not Detected	GC-FID	NEG	Not detected
Brodifacoum / Rat poison	Not Detected	HPLC	NEG	Not detected
Salicylate (Aspirin metabolites)	Not Detected	Trinder Reaction	NEG	Not detected — allergy documented

Substance / Test	Result	Method	Status	Reference Range
Fluoroacetate	Not Detected	<i>Ion-Selective Electrode</i>	NEG	<i>Not detected</i>

Envenomation screen (snake venom, scorpion venom, insect venom panel): Not indicated. No history of bite, sting, or environmental exposure to venom-producing organisms. This panel is available at the CGH Toxicology Laboratory on clinical request.

5.4 Heavy Metals Panel — Inductively Coupled Plasma Mass Spectrometry (ICP-MS)

A 12-element heavy metals panel was performed by ICP-MS, considered the analytical gold standard for trace element and heavy metal quantification in biological matrices. ICP-MS offers superior sensitivity (sub-ppb detection limits), multi-element capability, and high throughput. Results are presented in Table 5.

Table 6.

12-element heavy metals panel by Inductively Coupled Plasma Mass Spectrometry (ICP-MS). Reference ranges are for adult males. LOW = below lower limit of reference range.

Element (Symbol)	Result	Method	Status	Reference Range
Lead (Pb)	18 µg/dL	<i>ICP-MS</i>	NORMAL	< 25 µg/dL (<i>adult</i>)
Mercury (Hg)	4.2 µg/L	<i>ICP-MS</i>	NORMAL	< 10 µg/L
Arsenic (As)	12 µg/L	<i>ICP-MS</i>	NORMAL	< 35 µg/L
Cadmium (Cd)	0.8 µg/L	<i>ICP-MS</i>	NORMAL	< 5 µg/L
Chromium (Cr)	1.1 µg/L	<i>ICP-MS</i>	NORMAL	< 5 µg/L
Copper (Cu)	112 µg/dL	<i>ICP-MS</i>	NORMAL	70–140 µg/dL
Zinc (Zn)	68 µg/dL	<i>ICP-MS</i>	LOW	70–120 µg/dL — <i>Supplementation initiated</i>
Iron (Fe)	78 µg/dL	<i>ICP-MS</i>	NORMAL	60–170 µg/dL
Selenium (Se)	98 µg/L	<i>ICP-MS</i>	NORMAL	70–150 µg/L
Thallium (Tl)	Not detected	<i>ICP-MS</i>	NEG	< 0.5 µg/L
Manganese (Mn)	1.4 µg/L	<i>ICP-MS</i>	NORMAL	< 15 µg/L
Nickel (Ni)	2.1 µg/L	<i>ICP-MS</i>	NORMAL	< 8 µg/L

Clinical note: Mild zinc deficiency (68 µg/dL; reference: 70–120 µg/dL) was identified. Oral zinc sulfate 20 mg once daily was commenced on Day 6 and prescribed for 30 days post-discharge. All other elements were within their respective reference ranges. No evidence of toxic metal exposure was found.

6. DIAGNOSTIC IMAGING & MACHINE-BASED INVESTIGATIONS

6.1 12-Lead ECG

Equipment: Mortara Eli 250 12-Channel · **Dates:** 22 Apr 2026 & 30 Apr 2026

Admission (22 Apr): Sinus tachycardia, HR 98 bpm. PR interval 162 ms (normal). QRS 88 ms (normal). QTc 420 ms (normal). No ST-T changes. Normal electrical axis (+45°). Impression: Sinus tachycardia consistent with febrile illness.

Discharge (30 Apr): Normal sinus rhythm, HR 72 bpm. All intervals within normal limits. QTc 410 ms. No ischaemic changes. Impression: Normal ECG at discharge.

6.2 Chest X-Ray — PA View (Digital)

Equipment: Siemens Multix Impact DR (120 kVp/20 mAs) · **Dates:** 22 Apr 2026 & 29 Apr 2026

Admission (22 Apr): Mild homogeneous opacity right costophrenic angle consistent with small pleural effusion. Bilateral lung fields otherwise clear. Cardiac silhouette normal (CTR 0.48). No pneumonic consolidation, pneumothorax, or mediastinal widening. Impression: Mild right-sided pleural effusion — dengue-related capillary leak.

Follow-up (29 Apr): Right costophrenic angle completely cleared. Bilateral lung fields clear. No residual pleural effusion. Impression: Complete resolution of pleural effusion.

6.3 Abdominal Ultrasonography

Equipment: Samsung Medison RS85 — 3.5 MHz Convex Probe · **Dates:** 22 Apr 2026

Liver mildly enlarged (longitudinal diameter 16.2 cm). Hepatic echotexture mildly heterogeneous consistent with hepatitis. Spleen mildly enlarged (13.8 cm). Mild free fluid in perihepatic region and Morrison's pouch — estimated ~120 mL. Gallbladder wall mildly thickened (4.2 mm) with pericholecystic oedema. Both kidneys: normal size, shape and echotexture. No hydronephrosis. Impression: Hepatosplenomegaly, mild ascites, pericholecystic oedema — findings consistent with dengue haemorrhagic fever.

6.4 CT / MRI / 2D Echocardiogram / PFT

Not clinically indicated during this admission. Patient maintained haemodynamic stability and responded to conservative management throughout. CT imaging, MRI, echocardiography, and pulmonary function testing should be considered if neurological manifestations, cardiac symptoms, or persistent respiratory compromise are noted during follow-up.

7. THERAPEUTIC INTERVENTIONS

7.1 Procedures & Clinical Interventions

Table 7.
Chronological log of all clinical procedures and interventions performed during the admission.

Date	Procedure	Performed By	Outcome / Notes
22 Apr 2026	IV Cannulation — 18G Right forearm	Dr. R. Patil / Nr. D. Thorat	Uncomplicated
22 Apr 2026	Tourniquet test (Rumpel-Leede)	Dr. S. Nair	Positive — 22 petechiae in a 1-inch² area
22 Apr 2026	Phlebotomy — 5-tube baseline panel	Nr. D. Thorat	Uncomplicated. Cold chain maintained.
25 Apr 2026	Single-donor platelet concentrate transfusion — 1 ATD	Nr. S. Jadhav / Dr. R. Patil	No adverse transfusion reaction. Vitals stable throughout.
25 Apr 2026	IV cannula re-siting — Left forearm 18G	Nr. D. Thorat	Previous site: mild Grade I phlebitis. New site patent and functional.
26 Apr 2026	IV fluid rate de-escalation to 40 mL/hr	Dr. S. Nair	Oral fluid intake adequate. Clinical improvement noted.
28 Apr 2026	IV access removal — transition to full oral medications	Nr. L. Bai	Site inspected: clean, no signs of infection.

All procedures were performed by appropriately credentialed personnel. Informed verbal consent was obtained prior to each procedure and documented in the patient's clinical file. Procedure notes are filed in the medical records system under Patient ID CGH-2024-04891.

7.2 Pharmacotherapy — Discharge Prescription

The following medications were prescribed at discharge by the treating physician, Dr. Suresh Nair (MCI-234567), on 30 April 2026. All medicines were dispensed from the CGH Pharmacy prior to the patient's departure from the ward.

Table 8.
Discharge prescription — all medications prescribed for post-discharge management.

Medication (Generic Name)	Dosage / Frequency	Duration	Clinical Instructions
Tab. Paracetamol 500 mg	SOS (temp > 100°F only)	5 days	Do not exceed 4g/day total. AVOID Aspirin and all NSAIDs.
Tab. Metformin 500 mg	BD (after breakfast and dinner)	Indefinite — continuation	Type 2 DM management. Do not discontinue without physician review.
Tab. Amlodipine 5 mg	OD (after breakfast)	Indefinite — continuation	Hypertension management. Monitor BP daily. Target < 130/80 mmHg.
Tab. Vitamin C 500 mg	BD (after meals)	14 days	Immune recovery support.
Tab. Pantoprazole 40 mg	OD (before breakfast)	14 days	Gastric mucosal protection during recovery.
Tab. Zinc Sulfate 20 mg	OD (after food)	30 days	Mild zinc deficiency detected on ICP-MS heavy metals panel.
Syp. Papaya Leaf Extract	As per pharmacist-dispensed label	14 days	Standardised formulation only. Evidence for platelet recovery support.
ORS Sachets (WHO formula)	1 sachet in 1 litre water — PRN	14 days	Oral rehydration. Use if fluid intake appears inadequate.

7.3 Follow-up Schedule

Table 9.
Structured follow-up schedule post-discharge. Attendance at Day 7 and Day 14 appointments is clinically mandatory.

Appointment	Investigations Required	Location / Clinician	Timing
07 May 2026 (Day 7)	CBC — Platelet, WBC, Hb (mandatory)	Dr. S. Nair OPD, Room 12, CGH	09:00–11:00 hrs
14 May 2026 (Day 14)	CBC + LFT (SGPT, SGOT, Bilirubin) + RBS Fasting	Dr. S. Nair OPD, Room 12, CGH	09:00–11:00 hrs
30 May 2026 (Day 30)	HbA1c + Fasting Lipid Profile + Physician Review	Dr. S. Nair OPD, Room 12, CGH	09:00–11:00 hrs
If Symptoms Recur	IMMEDIATE Emergency Department Attendance	CGH Emergency — Ground Floor	24 × 7

8. CLINICAL TEAM — PERSONNEL INVOLVED IN CARE

Table 10.
All clinical and nursing personnel involved in the direct care of the patient during the admission period.

Role	Name	Registration No.	Qualification
Head of Department — General Medicine	Dr. Anjali Mehta	MCI-123456	MD (Medicine), MRCP (UK)
Treating Physician	Dr. Suresh Nair	MCI-234567	MD (Medicine), DNB
Haematologist (Consultant)	Dr. Priya Kulkarni	MCI-345678	MD (Pathology), DM (Haematology)
Radiologist (Consultant)	Dr. Vikram Joshi	MCI-456789	MD (Radiology), FRCR
Dietitian	Ms. Sneha Patwardhan	IDAI-78901	B.Sc. Nutrition, Certified Dietitian
Nursing In-charge — Ward B2	Sr. Lalitha Bai	MSNC-78901	B.Sc. Nursing
Staff Nurse	Nr. Deepa Thorat	MSNC-89012	GNM
Staff Nurse	Nr. Swati Jadhav	MSNC-90123	GNM
Resident Medical Officer	Dr. Rahul Patil	MCI-567890	MBBS (Intern — Year 2)

9. FINANCIAL STATEMENT — ITEMISED BILLING

Billing period: 22 April 2026, 09:14 hrs to 30 April 2026, 11:30 hrs. Insurance: Star Health Insurance — Policy No. SH-MH-2024-112233. Medi Assist TPA Pvt. Ltd. | Claim No. MA-2026-48912. All charges are per the City General Hospital tariff applicable at the time of service. GST is exempted for healthcare services as per Section 12(2) of the CGST Act 2017, Schedule II, Entry 74. All amounts in Indian Rupees (INR).

Table 11.
Itemised billing statement. Subtotal rows (grey) denote category groups.

Category	Item Description	Qty	Unit	Rate (₹)	Total (₹)
Accommodation					
	General Ward Bed (Block B2)	8	Days	800	6,400
Nursing					
	Nursing Care & Monitoring	8	Days	400	3,200
Consultation					
	Treating Physician (× 6 visits)	6	Visits	600	3,600
	Haematologist (× 2 visits)	2	Visits	800	1,600
	Radiologist (× 2 reports)	2	Reports	500	1,000
	Dietitian (× 1 visit)	1	Visit	300	300
Pharmacy — IV					
	NS 0.9% 500 mL	18	Bottles	85	1,530
	RL 500 mL	12	Bottles	90	1,080
	Paracetamol 1g / 100 mL	24	Vials	65	1,560
	Pantoprazole 40 mg	8	Vials	45	360
	Ondansetron 4 mg	6	Ampoules	35	210
Pharmacy — Oral					
	Metformin, Amlodipine, Vitamins, Zinc	1	Course	208	208
Blood Products					
	Platelet Concentrate — 1 ATD	1	Unit	3,200	3,200

Category	Item Description	Qty	Unit	Rate ()	Total ()
Diagnostics					
	CBC with Differential (× 5)	5	Tests	280	1,400
	LFT Panel (× 3)	3	Tests	450	1,350
	KFT Panel (× 3)	3	Tests	380	1,140
	Blood Sugar — RBS/FBS (× 5)	5	Tests	120	600
	HbA1c	1	Test	480	480
	NS1 Antigen (× 2)	2	Tests	800	1,600
	Dengue Serology IgG/IgM (× 2)	2	Tests	950	1,900
	Dengue PCR — RT-PCR	1	Test	2,200	2,200
	Electrolyte Panel (× 3)	3	Tests	320	960
	Coagulation Profile — PT/INR (× 2)	2	Tests	480	960
	CRP + ESR (× 2)	2	Tests	280	560
	LDH + Ferritin (× 2)	2	Tests	380	760
	Lipid Profile + TSH	2	Tests	770	770
	HIV / HBsAg / Anti-HCV Panel	1	Panel	680	680
	Blood Culture × 2 (aerobic)	1	Set	850	850
	Urine R/M + Culture (× 2)	2	Tests	180	360
	Stool Occult Blood	1	Test	150	150
	Toxicology Screen (DOA + BAC)	1	Panel	1,800	1,800
	Heavy Metals Panel — ICP-MS (12-element)	1	Panel	3,200	3,200
	Chest X-Ray PA (× 2)	2	Films	350	700
	Abdominal USG	1	Test	900	900
	12-Lead ECG (× 2)	2	Tests	200	400
Procedures					
	IV Cannulation (× 2 sites)	2	Procedures	150	300
	Platelet Transfusion — Administration	1	Episode	500	500
Miscellaneous					
	Admission & Registration	1	—	500	500
	Nursing Consumables & Disposables	1	Lot	750	750
	Discharge Report Generation	1	—	200	200
TOTAL AMOUNT PAYABLE — CITY GENERAL HOSPITAL					48,244

10. CLINICAL COURSE — DAY-BY-DAY NARRATIVE

The following section provides a chronological narrative of the patient's clinical course during the admission period. This handover document is intended for use by any subsequent treating physician, specialist, or emergency department receiving the patient in future. It should be read in conjunction with the serial laboratory data presented in Table 2 and the vital signs trend charts in Figures 1–4.

Day 1 — 22 Apr

Admitted OPD self-referral. 5-day history high-grade fever, retro-orbital headache, myalgia, nausea. NS1 Antigen positive. Admission CBC: WBC 3,200 cells/ μ L, Plt 65,000/ μ L (0.65 lakhs), Hb 13.2 g/dL. IV NS 0.9% commenced at 75 mL/hr. Paracetamol IV 1g TDS. Dengue protocol activated: 4-hourly vitals, CBC twice daily. Full baseline panel sent including dengue PCR, blood cultures $\times$ 2, LFT, KFT, electrolytes, coagulation, toxicology and heavy metals screens. CXR: mild right pleural effusion. USG: hepatosplenomegaly, mild ascites (~120 mL), pericholecystic oedema. ECG: sinus tachycardia 98 bpm.

Day 2 — 23 Apr

Fever spiking to 104.1°F. Platelet declining to 60,000/ μ L. IV Ondansetron 4 mg commenced for nausea. Oral intake poor. Family counselled on dengue progression and thrombocytopenia monitoring protocol.

Day 3 — 24 Apr

Platelet 52,000/ μ L — trend downward. Dengue IgG/IgM results: positive/positive — secondary dengue infection confirmed. Transaminitis: SGPT 88 U/L, SGOT 102 U/L. Diabetic diet commenced. Metformin continued. Haematologist consultation requested for Day 4.

Day 4 — 25 Apr

Platelet nadir: 38,000/ μ L. Haematologist Dr. Kulkarni reviewed — decision to administer prophylactic single-donor platelet concentrate (1 ATD). Transfusion administered over 60 minutes without adverse reaction. Post-transfusion platelet count: 54,000/ μ L. IV cannula re-sited (left forearm) due to Grade I phlebitis at original site. Dengue PCR result: DENV-2 serotype confirmed. IV fluids maintained at 75 mL/hr.

Day 5 — 26 Apr

Platelet 54,000/ μ L — early recovery. Fever reducing to 101.0°F. IV fluids de-escalated to 40 mL/hr. Oral intake improving. Rash fading. Patient more comfortable; ambulating short distances.

Day 6 — 27 Apr

Platelet 68,000/ μ L. Temperature 99.6°F. IV Ondansetron discontinued — tolerating oral medications and soft diet. Dietitian Ms. Patwardhan reviewed: diabetic soft diet, high fluid reinforced. Zinc deficiency noted on heavy metals panel — oral zinc supplementation commenced.

Day 7 — 28 Apr

Afebrile 98.9°F. Platelet 80,000/ μ L. IV access removed — fully transitioned to oral medications. Patient ambulatory, independent for all ADLs. General wellbeing markedly improved.

Day 8 — 29 Apr

Afebrile for 48 consecutive hours. Platelet 92,000/ μ L — satisfactory recovery trajectory. Repeat CXR: complete resolution of pleural effusion. ECG: normal sinus rhythm 72 bpm. Discharge criteria met. Discharge plan discussed with patient and family.

Discharge — 30 Apr

Final blood results reviewed. All improving satisfactorily. Structured 30-minute discharge education session conducted with patient and spouse (Nr. Lalitha Bai present as witness). Written discharge instructions, prescription, and follow-up card issued. All prescribed medications dispensed from CGH Pharmacy. Patient discharged ambulatory, in stable condition, accompanied by spouse.

11. DECLARATIONS, CERTIFICATIONS & ELECTRONIC SIGNATURES

11.1 Declaration by Treating Physician

I, Dr. Suresh Nair (Medical Registration No. MCI-234567), Treating Physician, hereby solemnly declare that I have personally examined and provided medical care to the above-named patient throughout the duration of this admission. All clinical information, investigation results, procedures performed, and medications administered as documented in this report are accurate and truthful to the best of my knowledge, professional assessment, and clinical judgement. The discharge diagnosis, condition at discharge, and post-discharge management plan are clinically appropriate and have been determined in the best interest of the patient. The patient and their authorised representative have been fully informed of the diagnosis, treatment provided, discharge plan, warning signs requiring emergency attendance, and the mandatory follow-up schedule.

_____	_____	_____
Dr. Suresh Nair (MCI-234567) Treating Physician	Date: 30 April 2026, 11:15 hrs	Sig ID: CGH-ESIG-MCI234567-20260430-1115 Auth: PIN + Biometric Fingerprint Verification

11.2 Countersignature — Head of Department

I, Dr. Anjali Mehta (Medical Registration No. MCI-123456), Head of Department of General Medicine, hereby countersign this discharge record and confirm that the clinical management of this patient has been reviewed by me and is in accordance with accepted standards of medical practice and the clinical protocols of City General Hospital. The treating team functioned under my departmental oversight as required by hospital policy.

_____	_____	_____
Dr. Anjali Mehta (MCI-123456) Head of Department — General Medicine	Date: 30 April 2026, 11:45 hrs	Sig ID: CGH-ESIG-MCI123456-20260430-1145 Auth: PIN + Biometric Fingerprint Verification

11.3 Hospital Certification

City General Hospital (Registration No. MH/HOS/2008/00341 | NABH Accreditation: NABH-H-0823) hereby certifies that this Patient Discharge Record has been generated by the hospital's certified Ward Management Information System. This document constitutes the official clinical discharge record of the patient named herein. Patient data has been processed in accordance with the Digital Personal Data Protection Act, 2023 (Consent Reference: CGPH-CONS-20260422-04891). This record shall be retained in accordance with the National Medical Commission guidelines and applicable Indian law.

_____	_____	_____
City General Hospital — Authorised Signatory Medical Superintendent / Authorised Representative	Date: 30 April 2026	Sig ID: CGH-SEAL-20260430 Auth: Official Hospital Digital Seal

11.4 Patient / Authorised Representative Acknowledgement

I, Rajesh Kumar Sharma, and/or my authorised representative Priya Sharma (Spouse), hereby acknowledge receipt of this Patient Discharge Record and confirm that I have received and understood the discharge instructions, prescribed medications, dietary restrictions, warning signs requiring emergency attendance, and the structured follow-up schedule, as explained by the treating physician and nursing team during the 30-minute structured discharge education session conducted on 30 April 2026. I consent to City General Hospital retaining my medical record in accordance with applicable law and the hospital's data protection policy under the Digital Personal Data Protection Act, 2023.

_____	_____	_____
Rajesh Kumar Sharma / Priya Sharma (Spouse) Patient / Authorised Representative	Date: 30 April 2026, 11:30 hrs Witness: Sr. Lalitha Bai (MSNC-78901)	Sig ID: CGH-PAT-ACK-20260430-04891 Auth: Patient Signature / Thumbprint Witnessed

12. DOCUMENT VERIFICATION & REGULATORY DISCLAIMERS



12.1 Document Integrity — QR Code Verification

This report carries a tamper-evident QR code encoding the Report ID, Patient ID, UHID, discharge date, and a cryptographic document hash (HMAC-SHA256). Scanning the QR code resolves to the hospital's secure document verification portal, which confirms the authenticity and integrity of the report. Any post-generation alteration to the printed document will cause verification to fail. This mechanism is compliant with the Information Technology Act, 2000 (as amended) and the Digital Personal Data Protection Act, 2023.

Report ID: CGH-DR-2026-04891-0430 · **Patient ID:** CGH-2024-04891 · **UHID:** UHID-MH-2024-948271 · **Discharge:** 30 April 2026 · **Generated:** 10 May 2026

12.2 Regulatory Disclaimers

Confidentiality.	This document constitutes a confidential medical record pertaining to the named patient. It is intended solely for the use of the patient, their authorised representatives, and healthcare professionals directly involved in the patient's ongoing care. Unauthorised disclosure, reproduction, or distribution of this document is prohibited under the Digital Personal Data Protection Act, 2023, and the Information Technology Act, 2000 (as amended).
Clinical Accuracy.	All clinical information has been recorded in good faith based on observations, investigations, and professional assessments made during the admission period. City General Hospital and its staff shall not be held liable for outcomes arising from reliance on this document without current clinical re-evaluation by a qualified healthcare professional.
Toxicology & Sensitive Results.	Toxicology, drug of abuse panel, and blood alcohol content results are disclosed only to the treating physician and the patient. Disclosure to third parties requires written patient consent or a valid court order under applicable Indian law, including the Narcotic Drugs and Psychotropic Substances (NDPS) Act 1985 where relevant.
Laboratory & Imaging.	Reference ranges are those established by CGH Pathology Laboratory (NABL Accredited) and may differ from ranges used by other institutions. Imaging and ECG reports represent the professional opinion of CGH specialists at the time of reporting. Clinical correlation is mandatory.
Billing.	All charges are per the hospital tariff applicable at time of service. GST is exempted for healthcare services under the CGST Act 2017. Billing disputes must be raised in writing to the Billing Department within 30 days of discharge.
Data Protection.	This patient's personal and health data is retained by City General Hospital in accordance with the National Medical Commission (NMC) guidelines (minimum 3-year retention for clinical records) and the Digital Personal Data Protection Act, 2023. The patient (Data Principal) may exercise their rights of access, correction, and erasure by submitting a formal written request to the Medical Records Department, quoting their Patient ID and UHID.
Limitation of Liability.	This discharge record is provided for clinical continuity purposes only. City General Hospital makes no warranty, express or implied, that the information contained herein is sufficient for clinical decision-making without direct examination of the patient by a qualified physician.

City General Hospital · 14, Civil Lines, Pune – 411 001, Maharashtra, India · Tel: +91-20-2612 3456 · NABH Accreditation: NABH-H-0823 · NABL Laboratory: NABL-2024-00781

— END OF CLINICAL DISCHARGE SUMMARY REPORT —